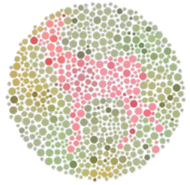
Hue-man Rights in Data Visualisation

How colour theory, and colour-blind friendliness all *shade* your data.

Jia Wei TEH Heidelberg University, IMPRS-HD jiaweiteh.astro@gmail.com

What we'll cover



01

How We See Colour

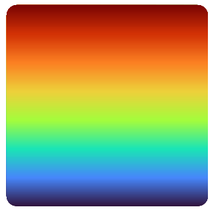
...and how we don't | perception & colour-blindness



02

When Colourmaps Lie

The good, the bad & the misleading



03

Breaking the Rules

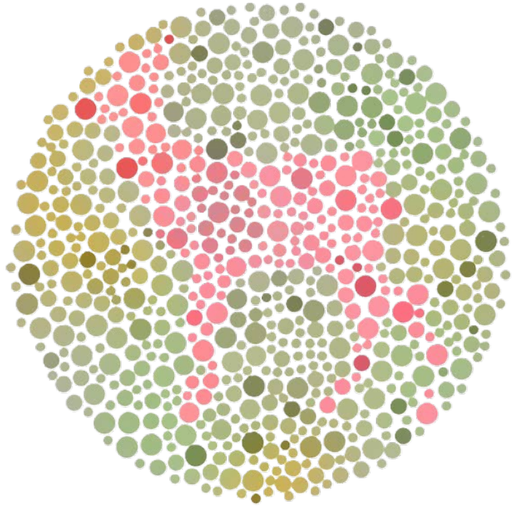
When rainbow maps actually help

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DISCLAIMER

Not a Doctor.med

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01

PART ONE

How We See Colour

...and how we don’t.

01 How We See Colour

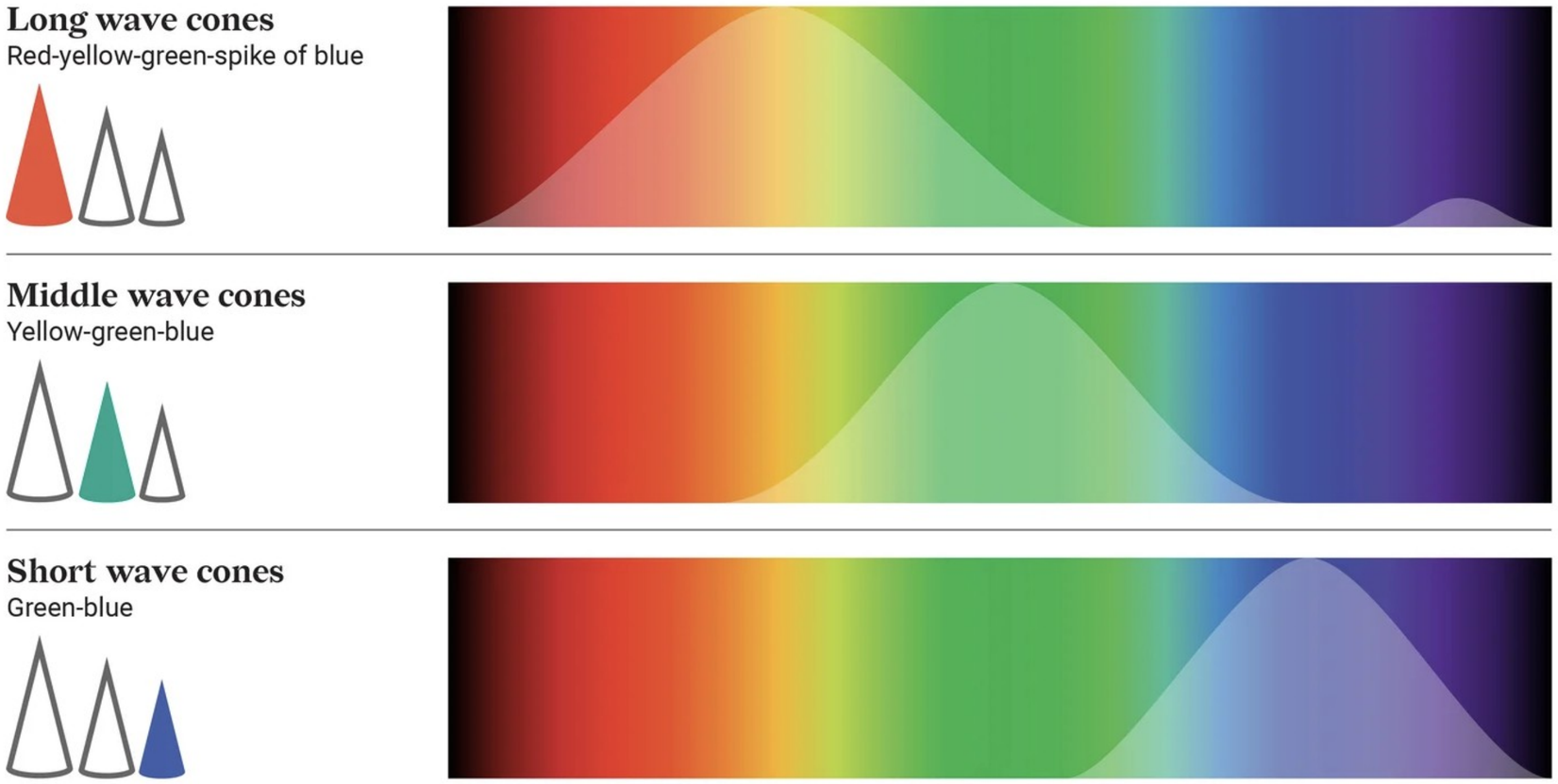
02 When Colourmaps Lie

03 Breaking the Rules

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How our eyes see colour

How three cone types create millions of colours.



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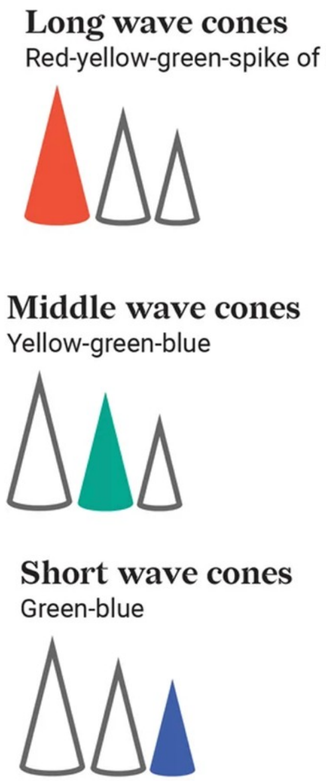
Your retina does colour maths

Three cones become three channels: red–green, blue–yellow, light–dark.

WHY MATH?

The retina transmits the **cmobination**, far more efficient than three near-identical signals. Every colour is a point in this 3-axis space (RG, BY, L)

It's why **"reddish-green"** and **"yellowish-blue"** don't exist!



R–G

Luminance

B–Y

See Hering (1878); Hurvich & Jameson (1957)

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Colour isn’t universal

Even people with ‘normal vision’ perceive/name colour differently.

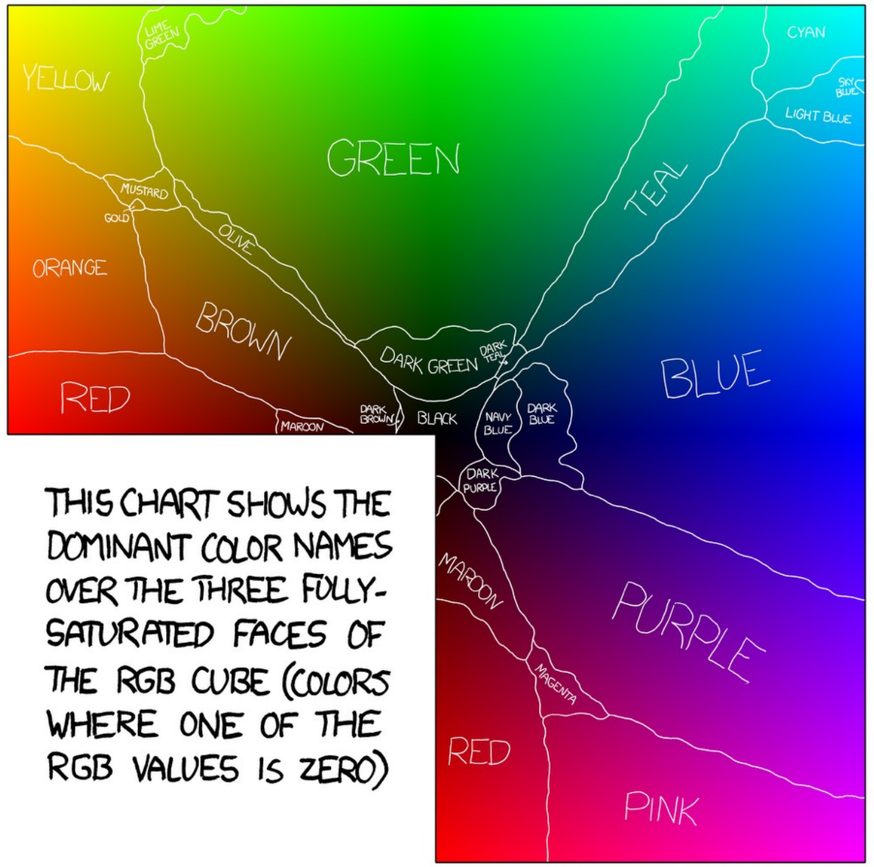
THE XKCD COLOUR SURVEY

Each participant is given a colour:



and asked to name it.

Here are the results from 200k+ volunteers:



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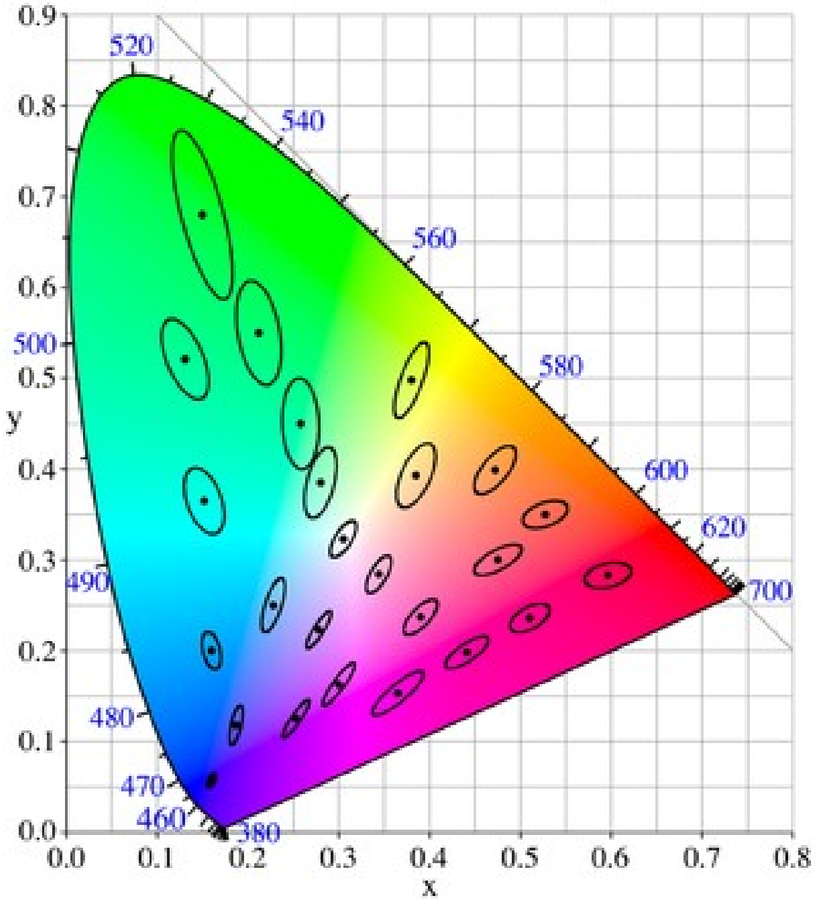
Colour isn’t universal

Even people with ‘normal vision’ perceive colour differently.

THE CIE COLOUR SPACE

These **Ellipses**, varying wildly in size and orientation, are examples of areas we **can’t tell colours apart.**

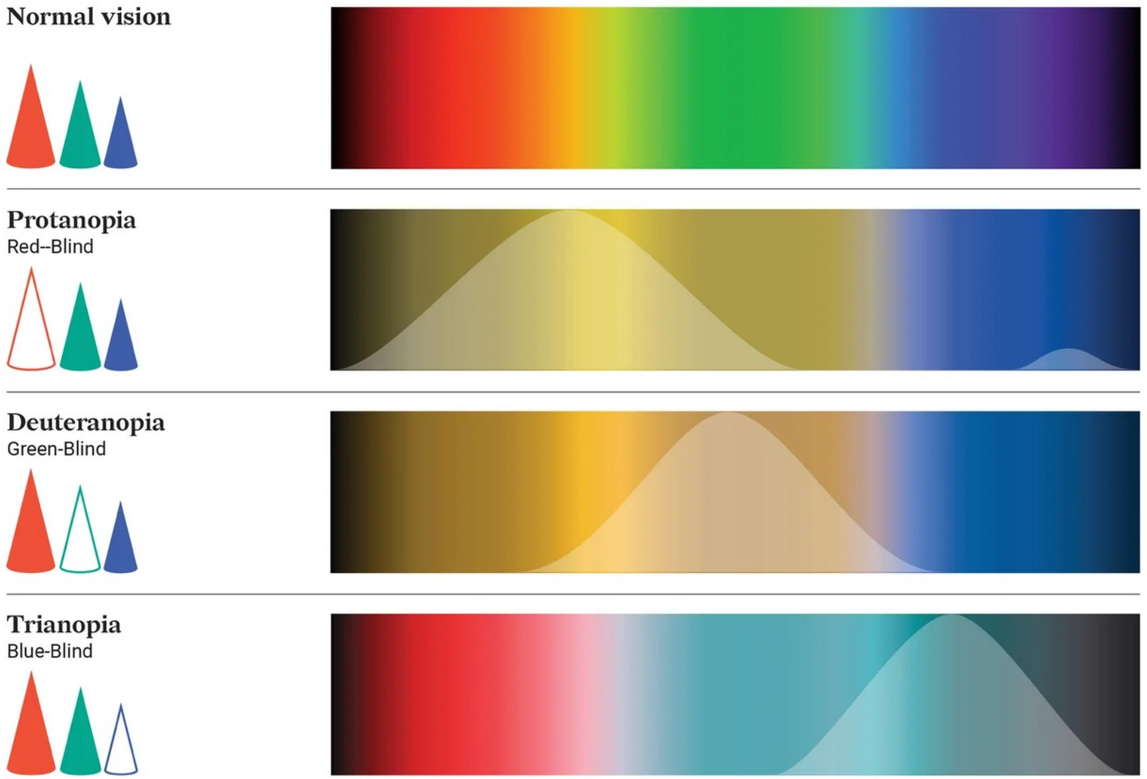
We’re **bad at discriminating greens** (big ellipses) but **good at blues** (small ellipses).



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Colour-blindness in a nutshell

What happens when cone cells don’t work.

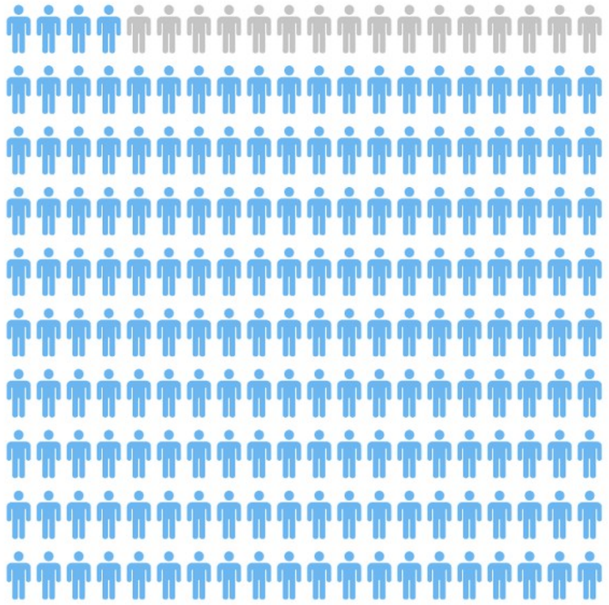


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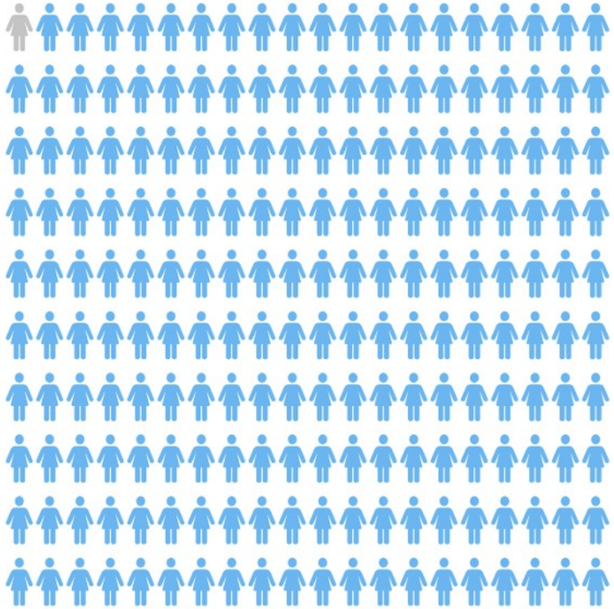
Why should we care?

You probably know **at least 13** colour blind people

(this could be your reviewer)



16 out of 200 men



1 out of 200 women

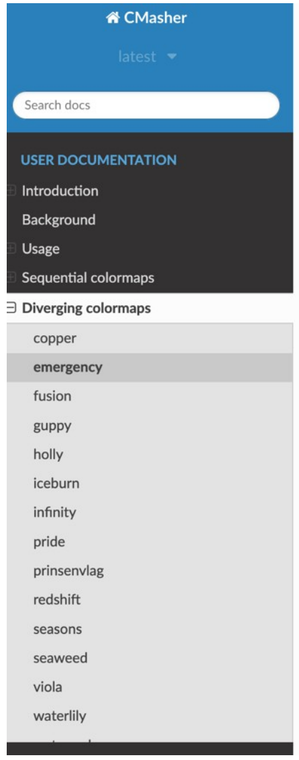
if humans didn't convince you enough...

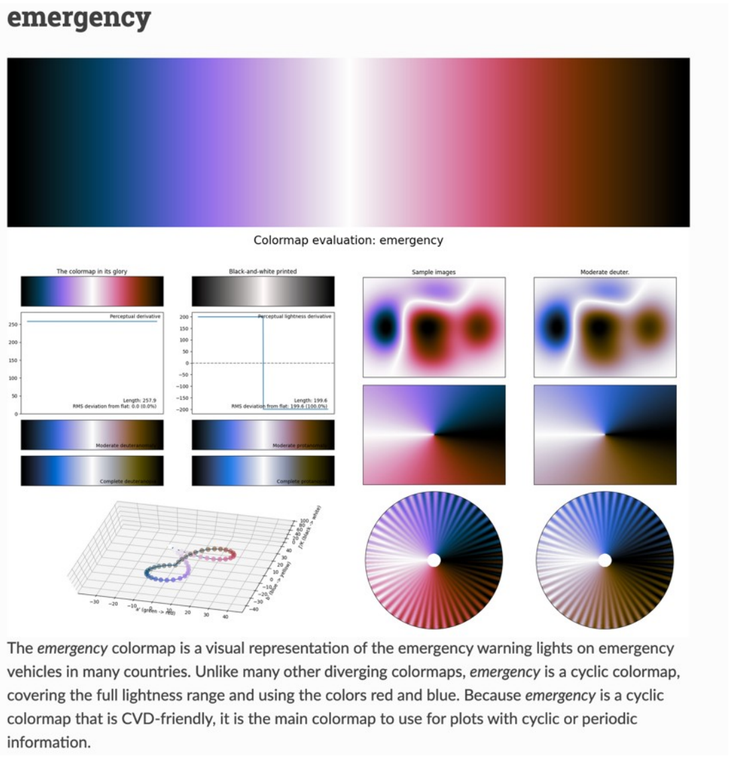
100 out of 100 dogs

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S O …

Use colour-blind-friendly palettes



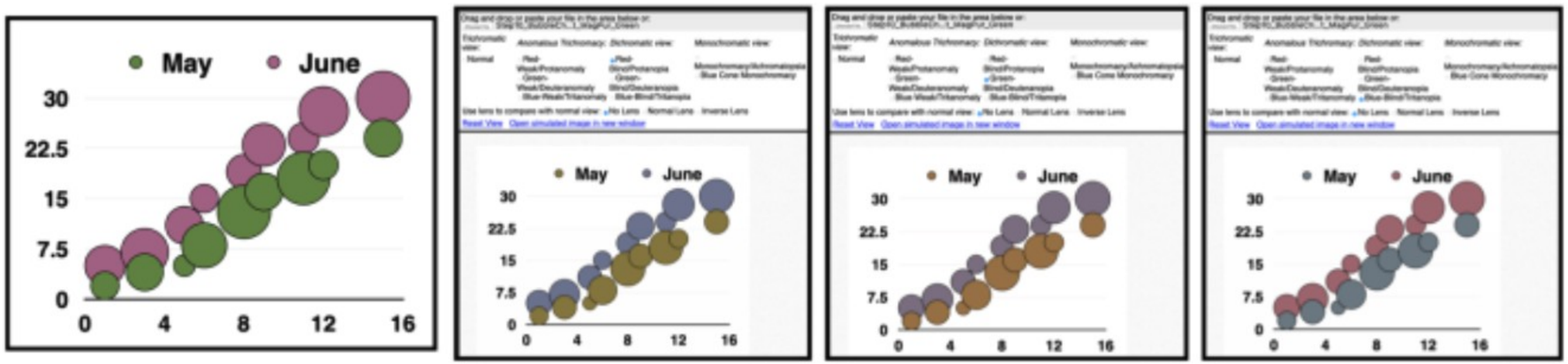


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S O …

Try colour-blind simulators

CoBliS — free online.



Normal Vision

Protanopia
Loss of Red Cone

Deuteranopia
Loss of Green Cone

Tritanopia
Loss of Blue Cone

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TAKE-HOME 01

Colour perception is unique.
Be inclusive!

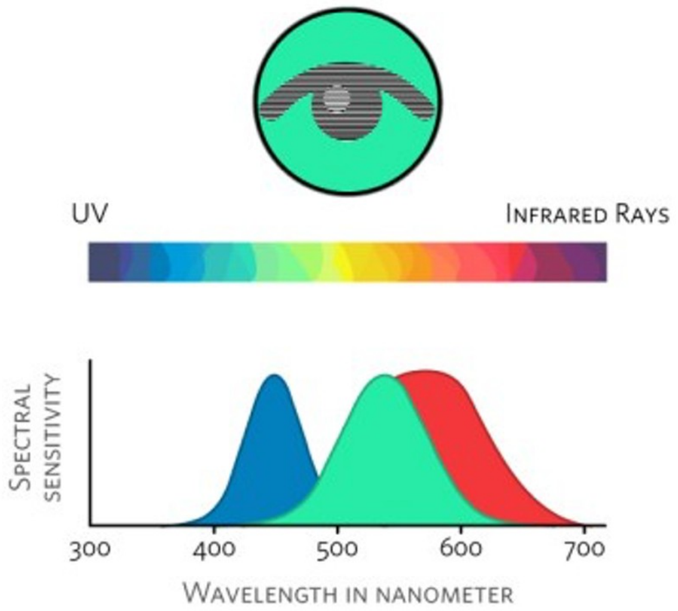


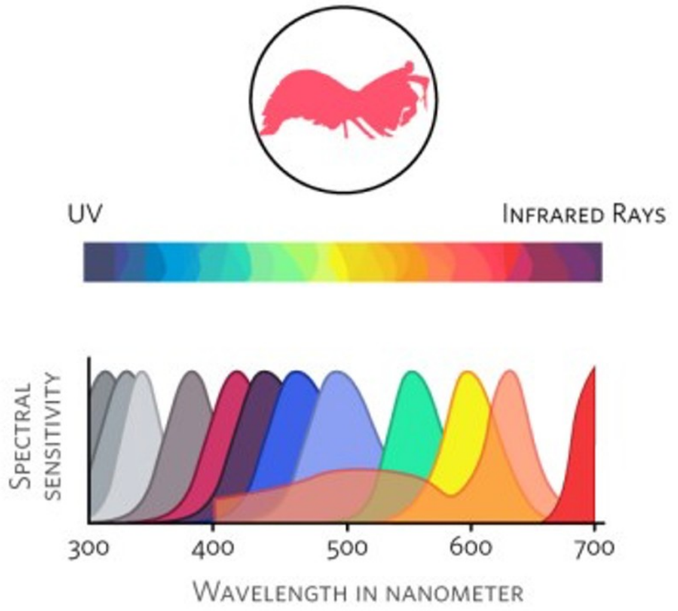
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TAKE-HOME 1.5

Mantis shrimp have up to 12 receptors

Imagine if their slides only worked for shrimps!





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02

PART TWO

When Colourmaps Lie

01 How We See Colour

02 When Colourmaps Lie

03 Breaking the Rules

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Human eyes work in logarithms

The Weber-Fechner brightness-perception.



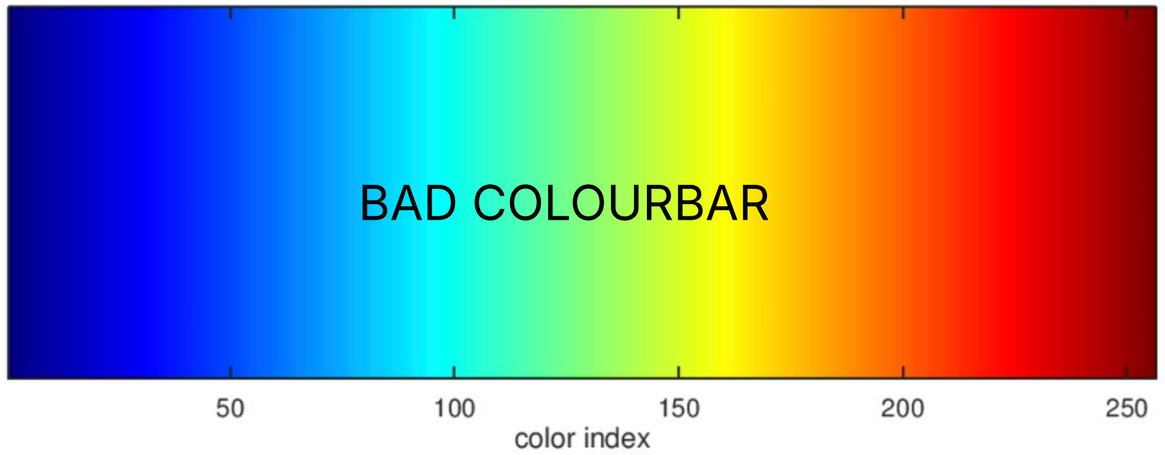
WE SEE ACROSS 14 ORDERS OF MAGNITUDE

We perceive brightness difference as ratios. Doubling light feels the same going from 1 to 2, or 100 to 200.

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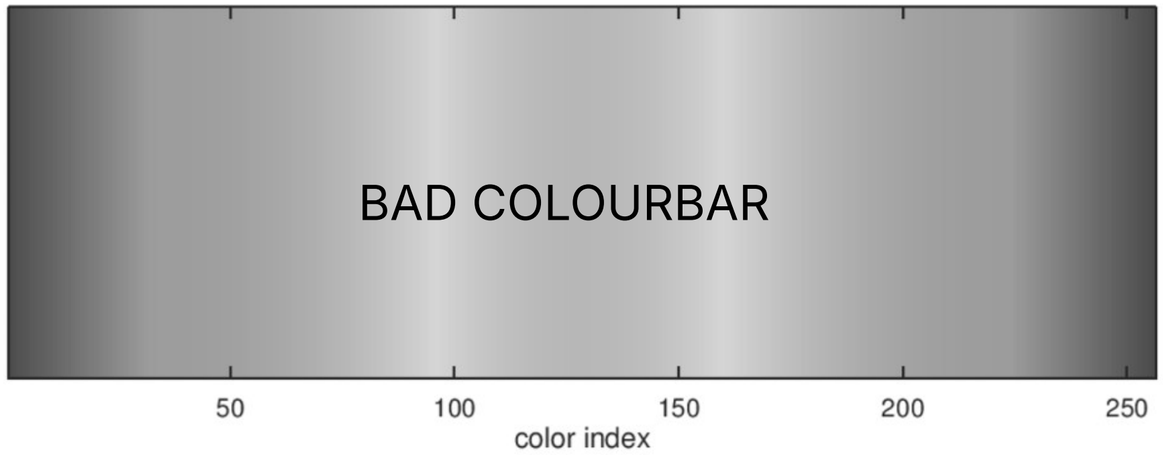
Perception uniformity

A 10% change in colour is a 10% change in data



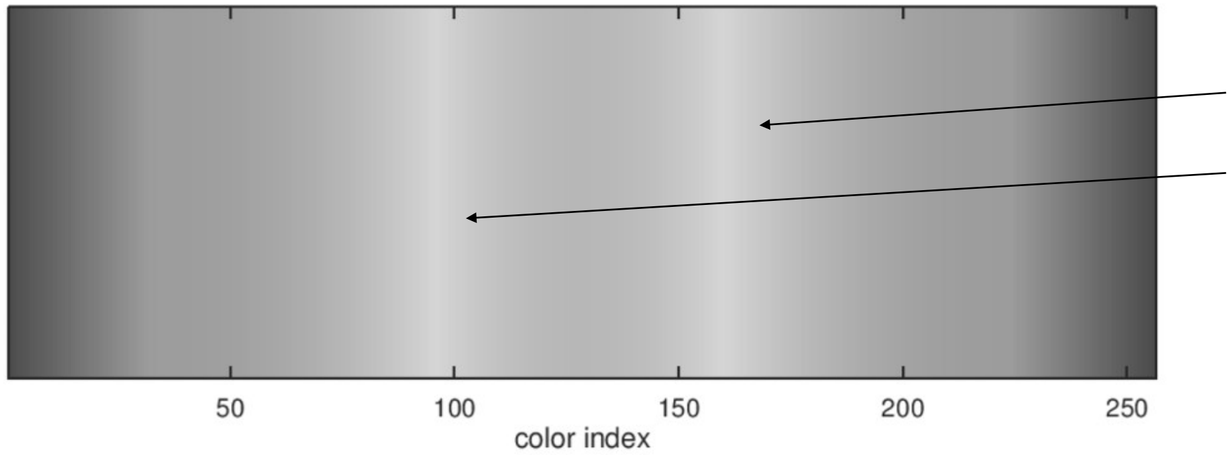
Perception uniformity

A 10% change in colour is a 10% change in data



Perception uniformity

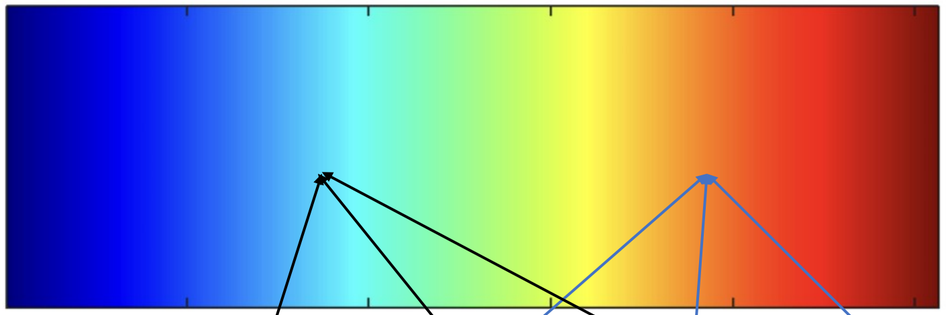
A 10% change in colour is a 10% change in data



Features of ***colourmap***, not ***data***

Perceptual uniformity, measured

Colour distance? What are those?

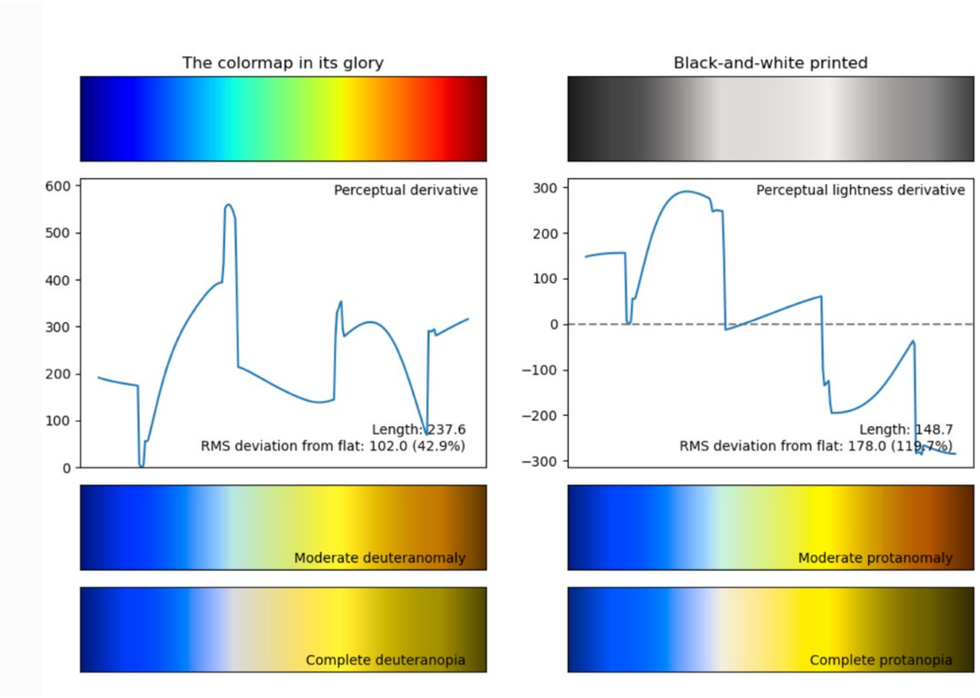


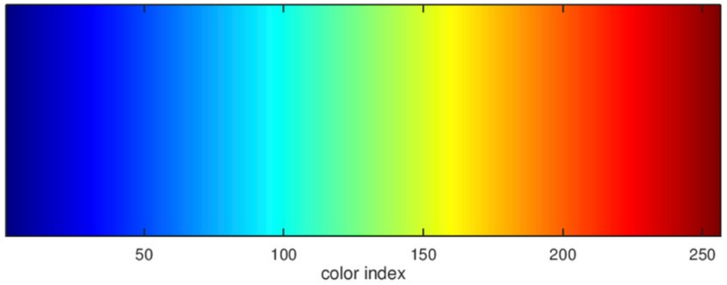
$$\text{distance} = \sqrt{(R_2 - R_1)^2 + (G_2 - G_1)^2 + (B_2 - B_1)^2}.$$

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Perceptual uniformity, measured

Colour distance? What are those?



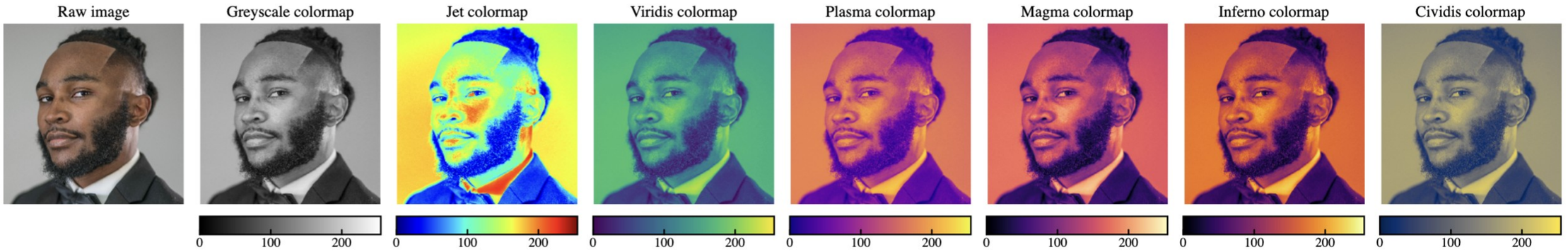


$$\text{distance} = \sqrt{(R_2 - R_1)^2 + (G_2 - G_1)^2 + (B_2 - B_1)^2}.$$

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The face test

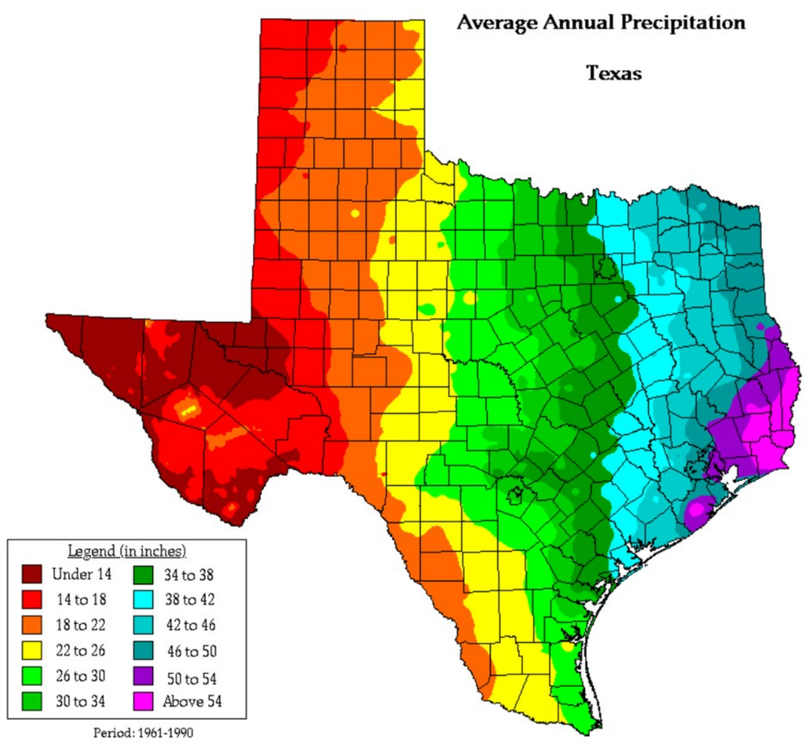
A natural-looking face is a reasonably uniform colourmap.

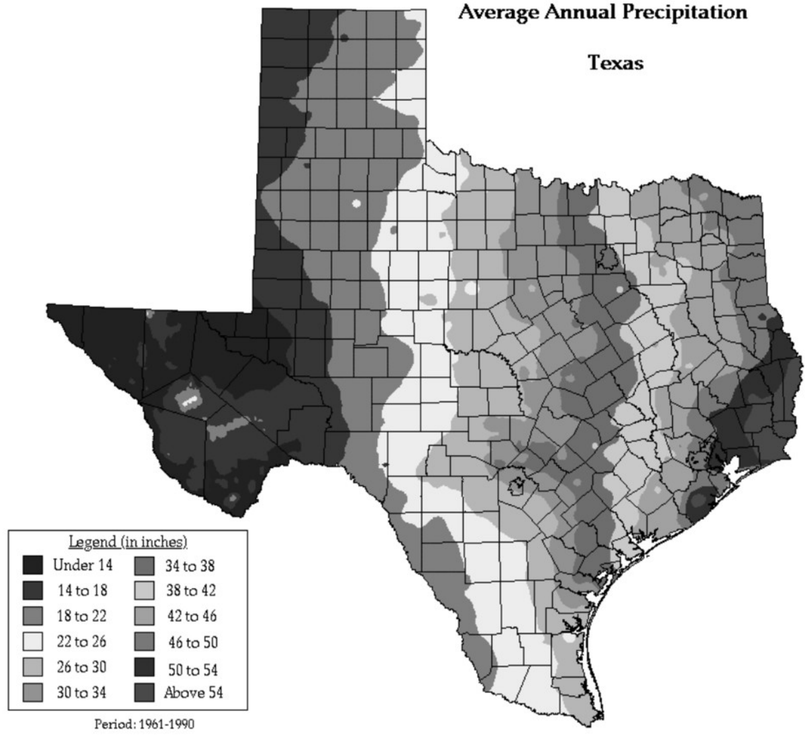


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Colour qualifies, the legend quantifies

A good colourbar should convey the trend even in grayscale

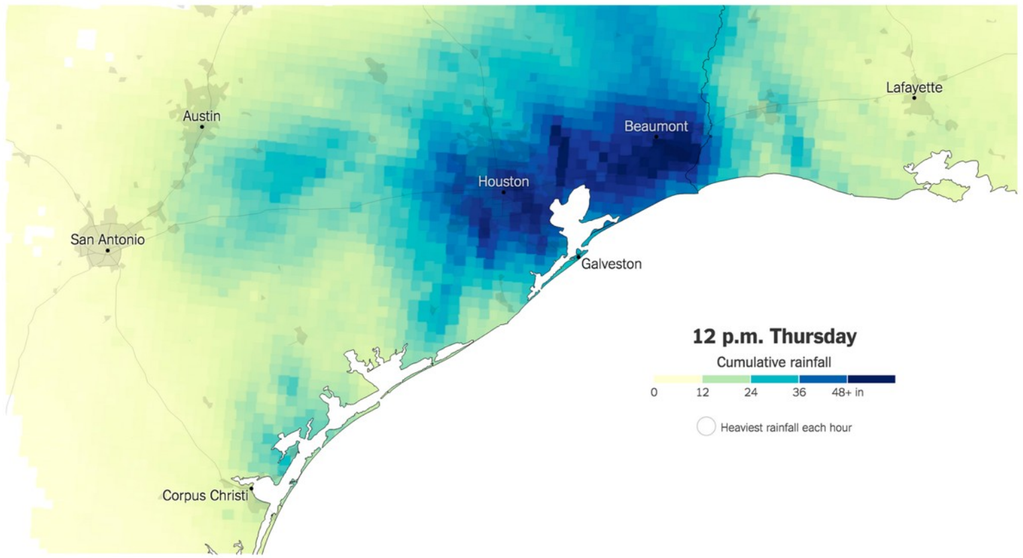


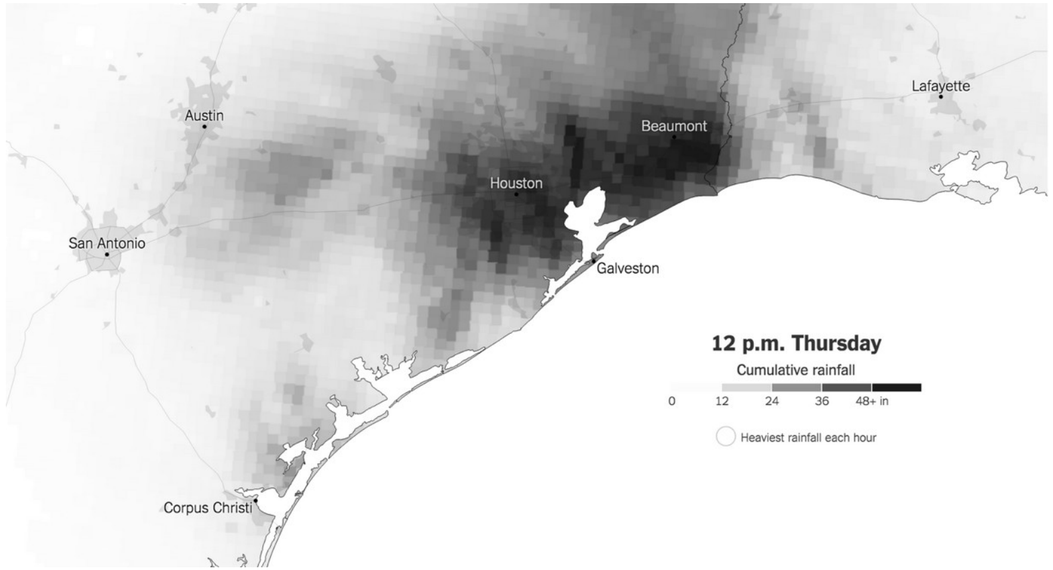


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Colour qualifies, the legend quantifies

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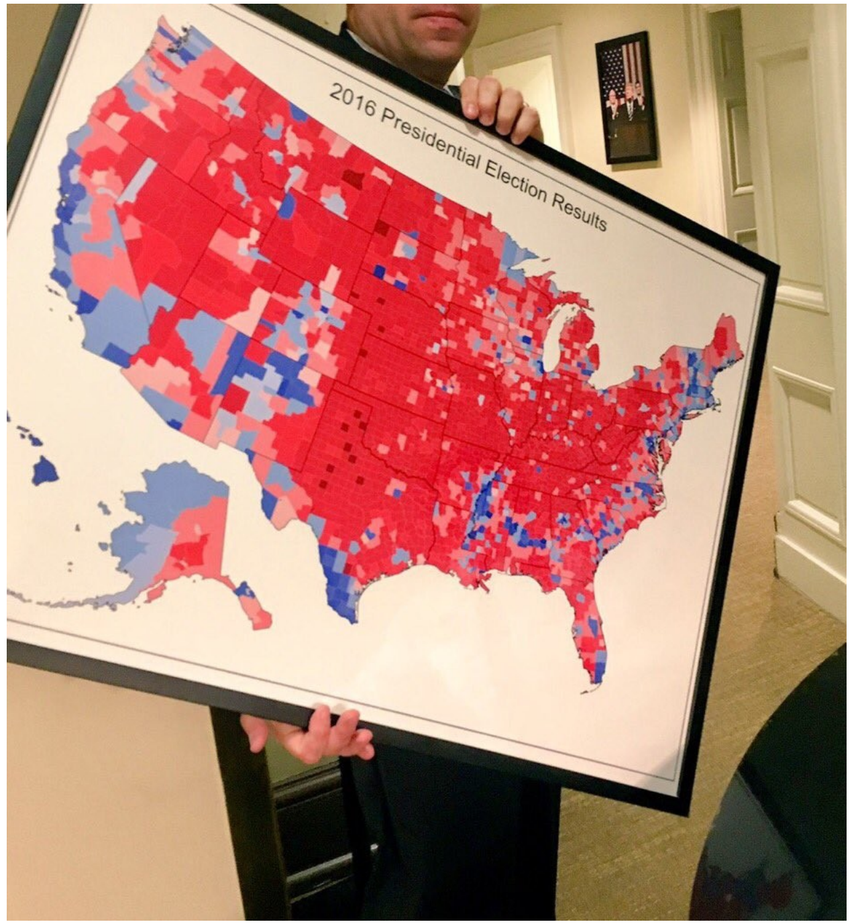




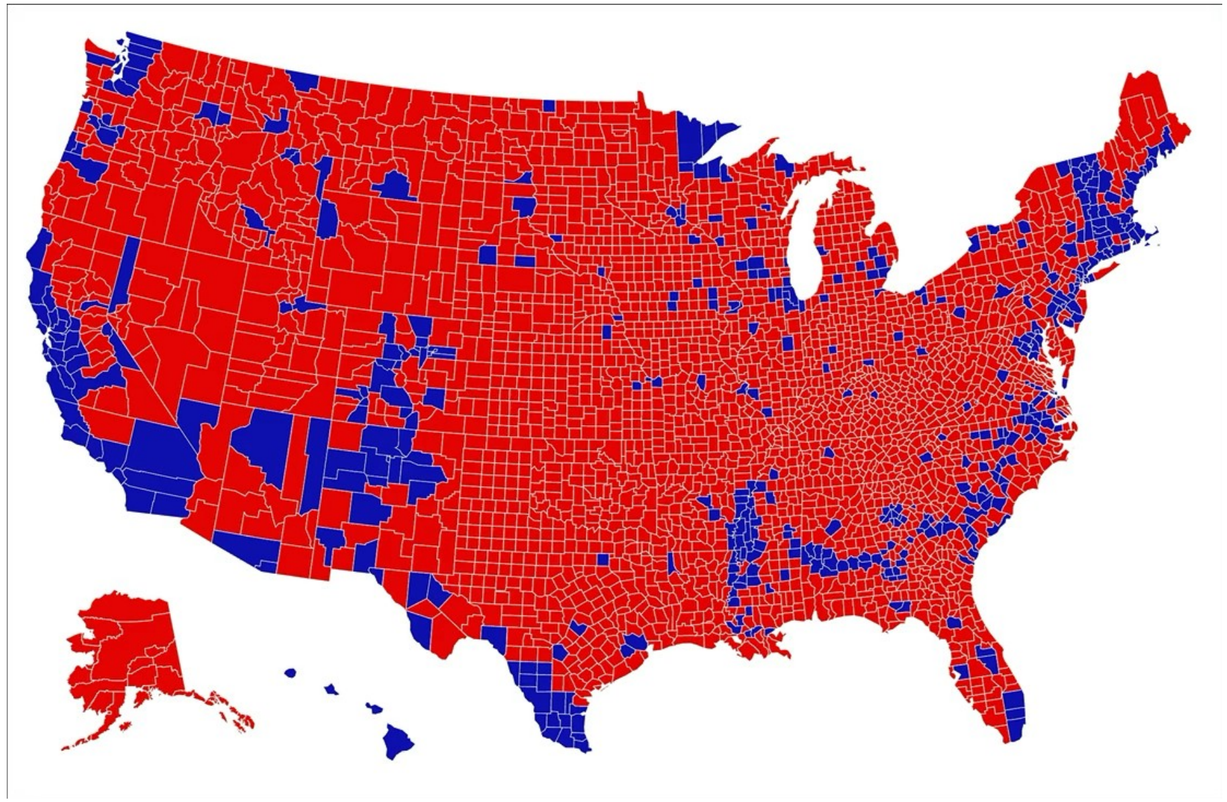
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Also important: the correct data normalisation

Weight is important to provide information on proportion and perspective



Coloured by county



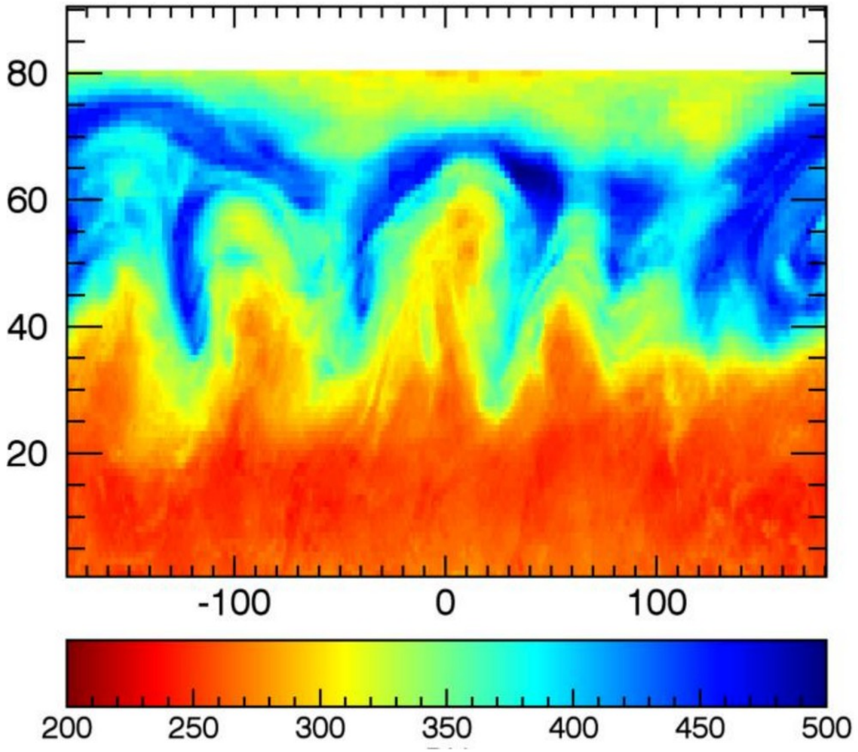
Coloured by county, weighted by population density

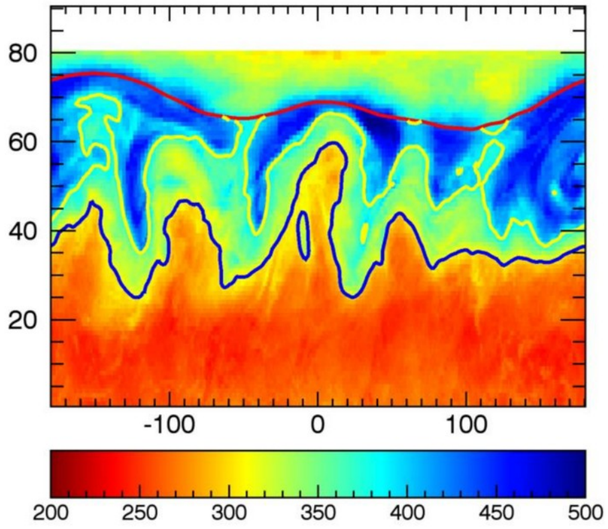
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Perceptual uniformity: case study of ozone concentrations.

Two layers? Three layers? The importance of colourbar.

Based on this map, there are three zones of intense concentration change. One can identify a polar front (red), a mid-latitudinal front (yellow) and a sub-polar front (blue).





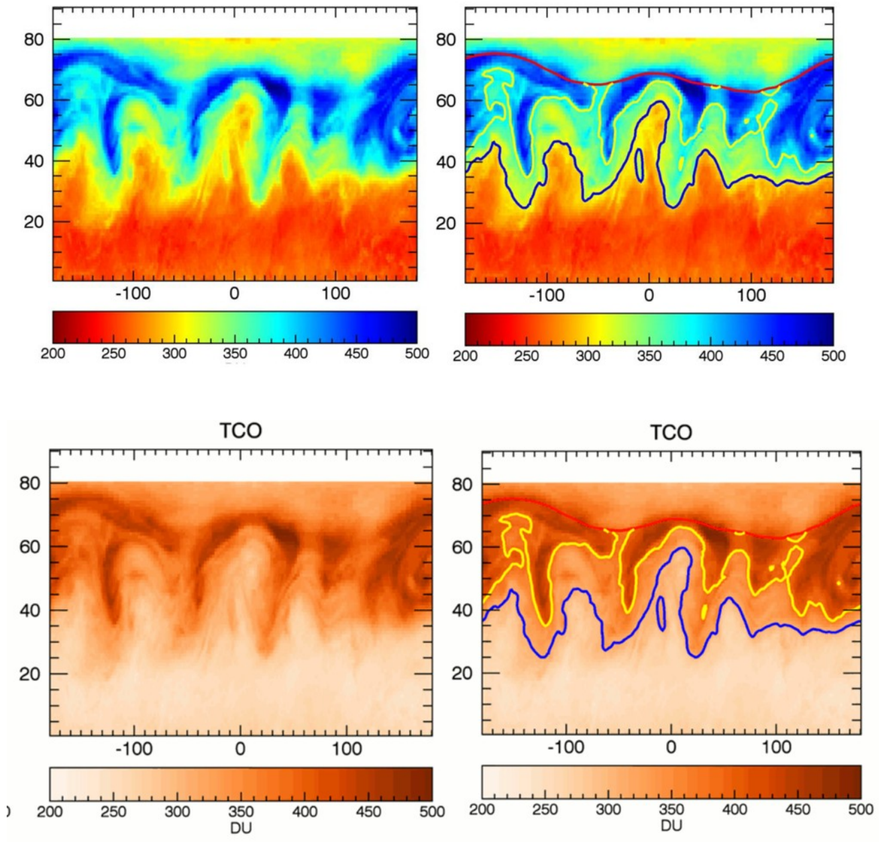
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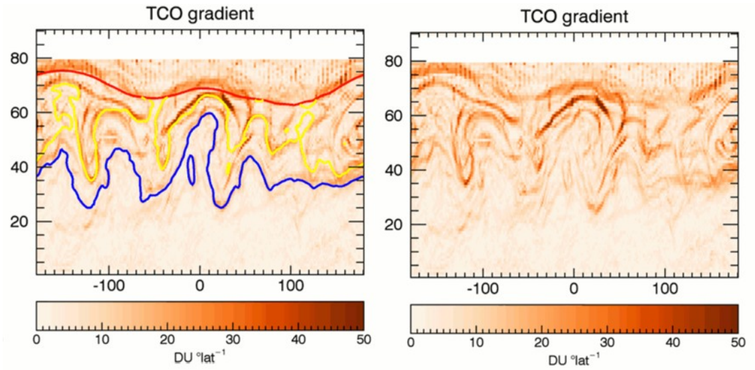
Perceptual uniformity: case study of ozone concentrations.

Two layers? Three layers? The importance of colourbar.

With a uniform colourbar, these ‘features’ disappear!

The abrupt change is an artifact of a non-uniform colourbar.



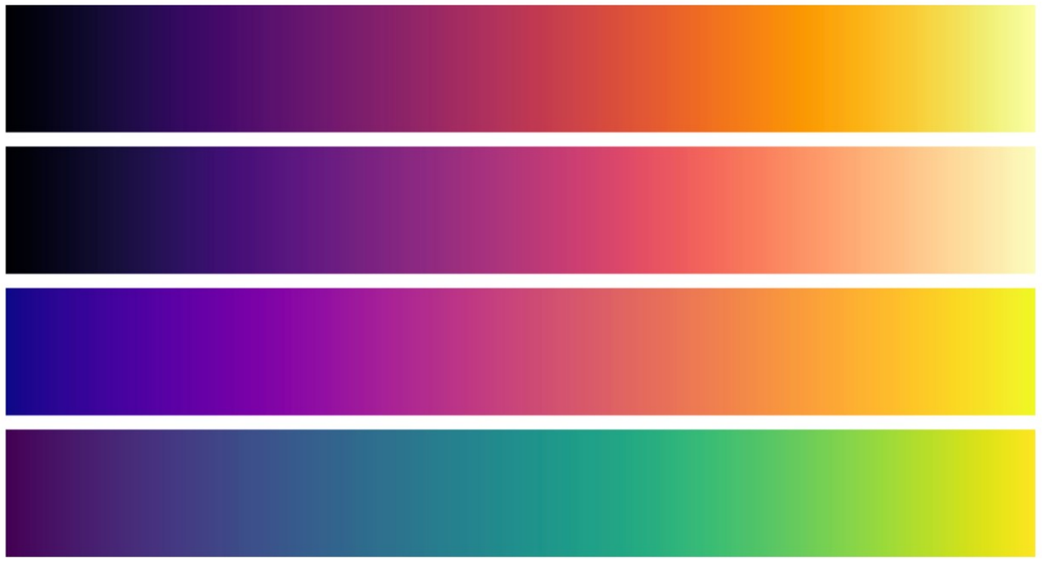


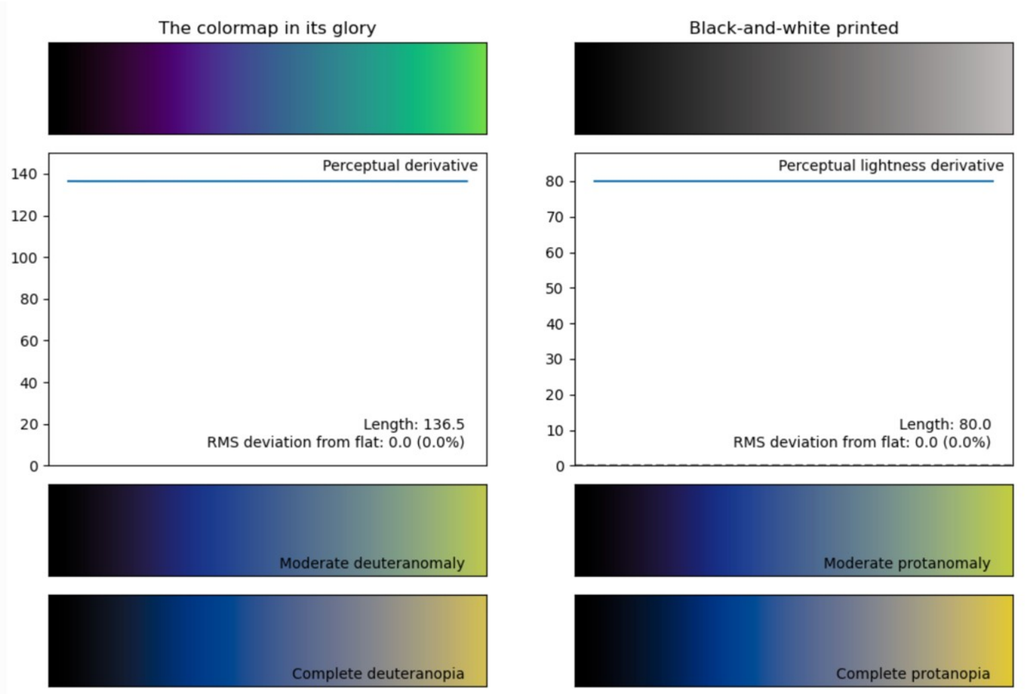
Jia Wei TEH Heidelberg University, IMPRS-HD jiaweiteh.astro@gmail.com

THE SOLUTION

Use perceptually uniform colourmaps

Inferno, Magma, Plasma, Viridis

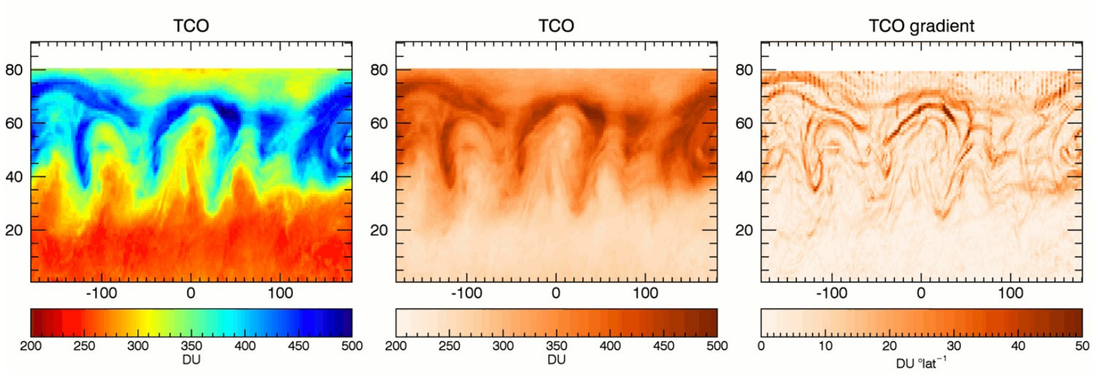




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TAKE-HOME 02

Perceptual uniformity carries accurate data.



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03

PART THREE

Breaking the Rules

01 How We See Colour

02 When Colourmaps Lie

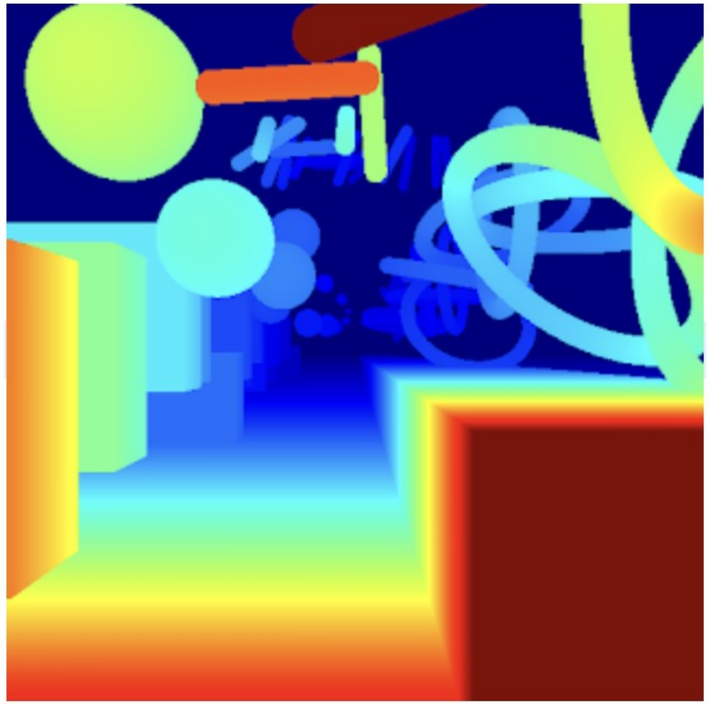
03 **Breaking the Rules**

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Every cloud has a silver lining...

Their striking features give rainbow maps decent depth perception.





Jet

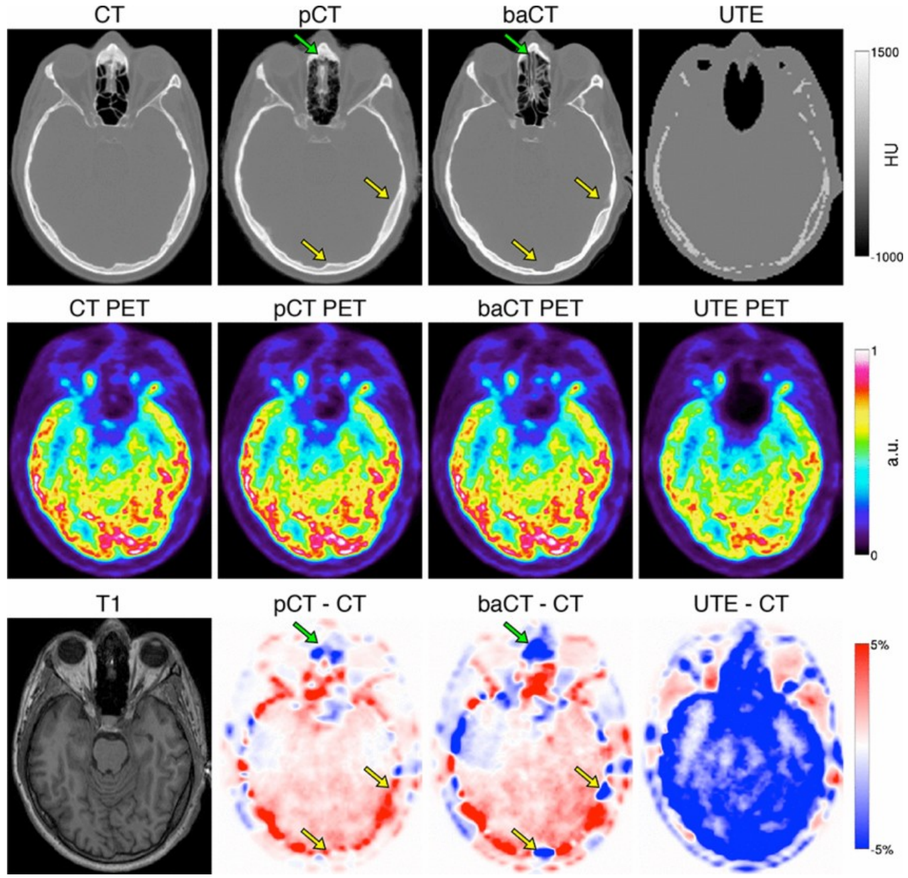


Inferno

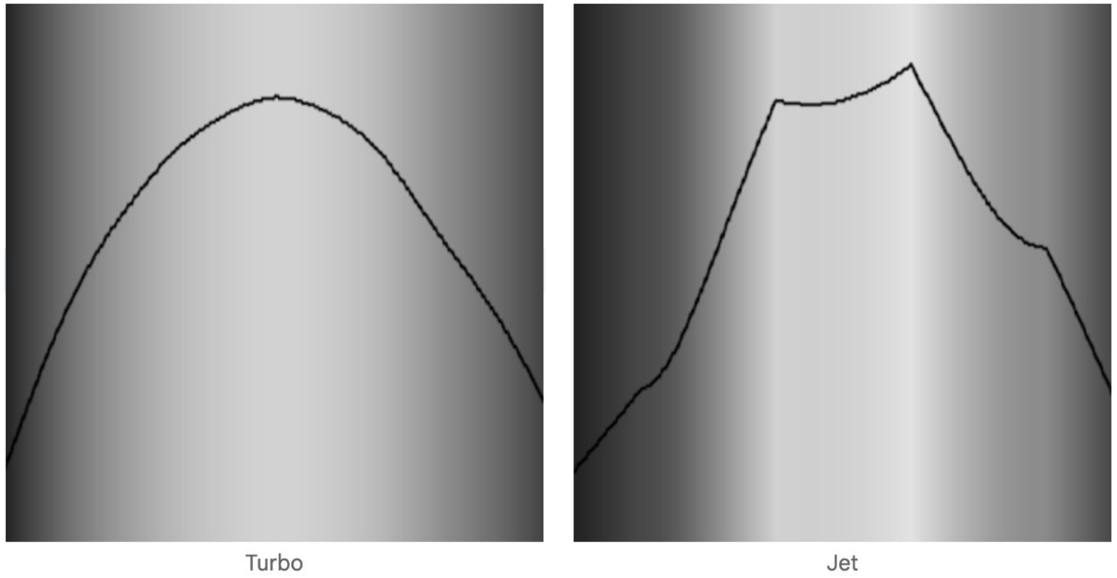
Jia Wei TEH Heidelberg University, IMPRS-HD jiaweiteh.astro@gmail.com

The upside of inaccurate maps

High-contrast maps reveal small details.

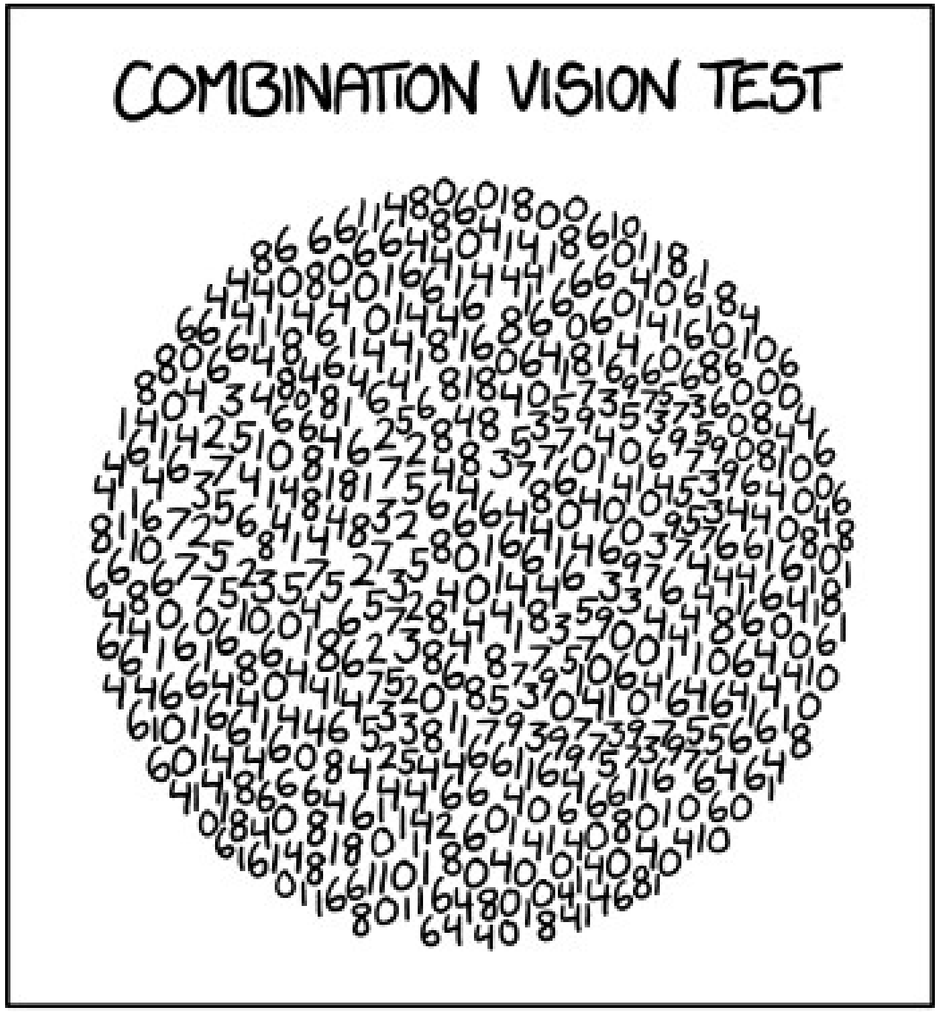






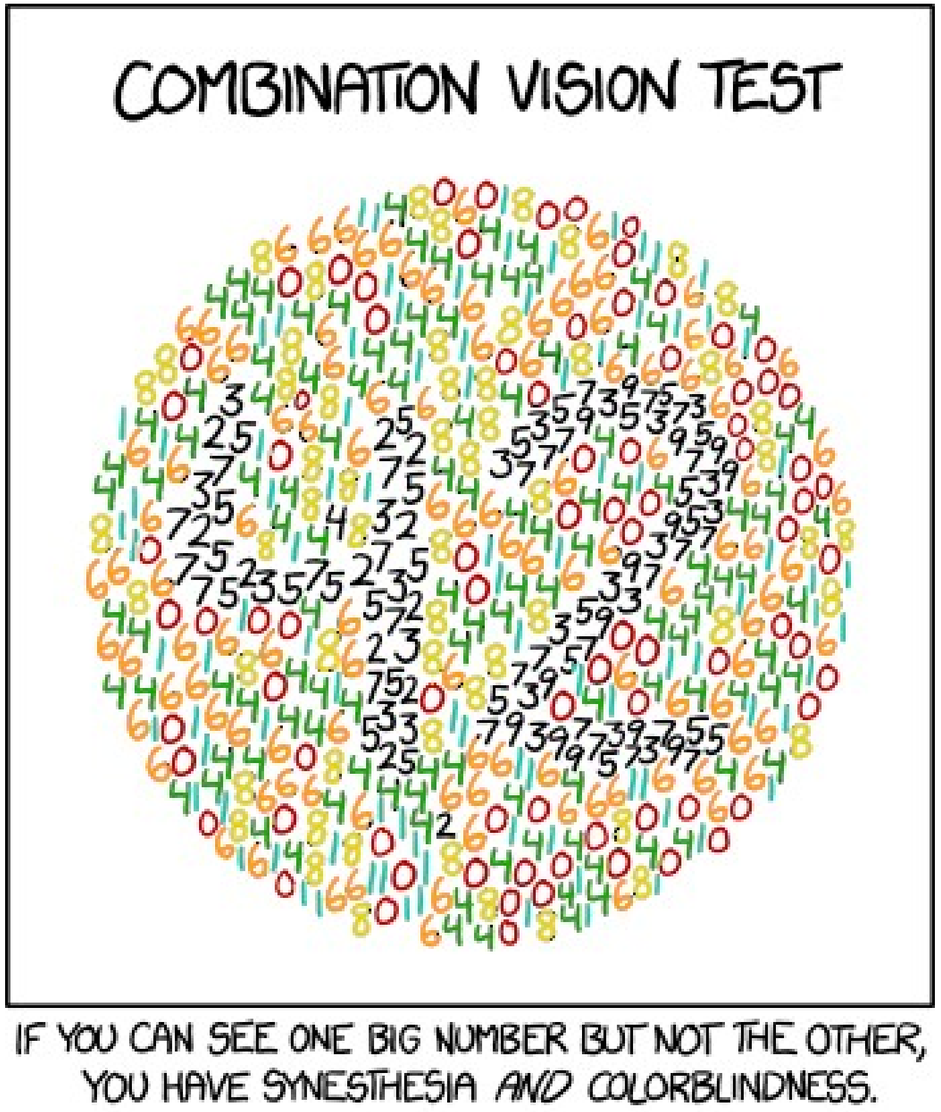
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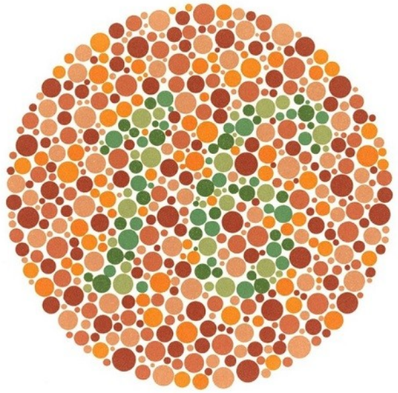
Thought experiment



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Thought experiment

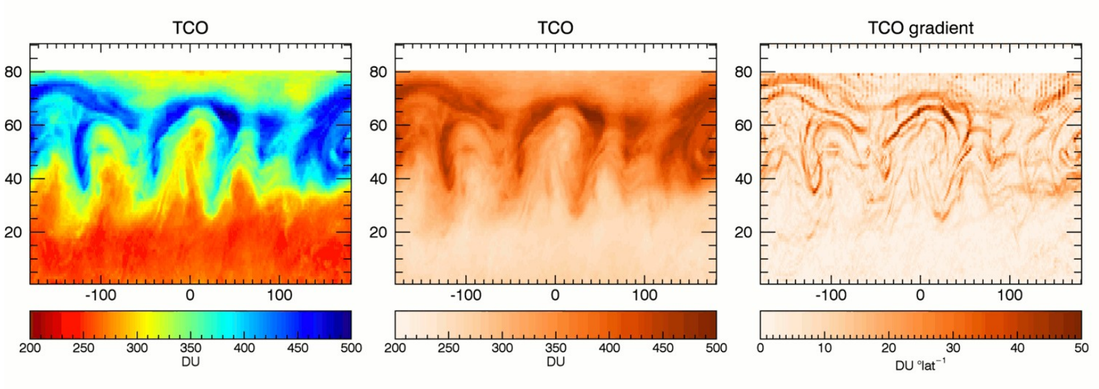


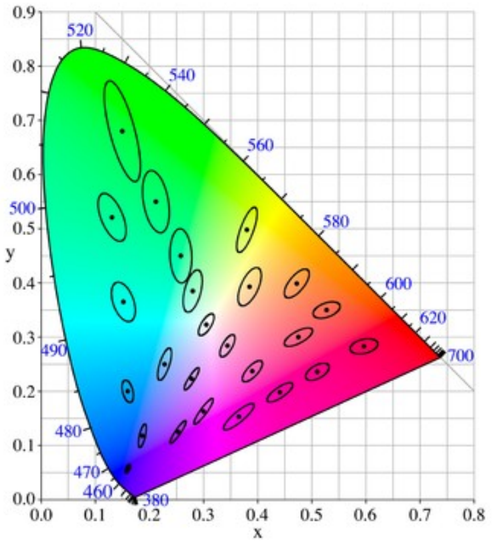


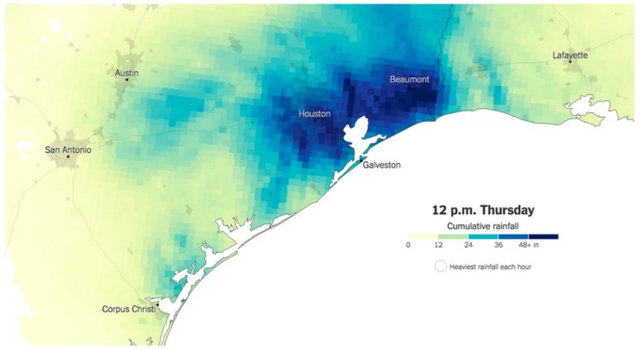
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Thank you

Stay inclusive and uniform!







Jia Wei TEH Heidelberg University, IMPRS-HD jiaweiteh.astro@gmail.com